

Hariprakash Karthikeyan

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EDUCATION

University of Rochester

MS, Data Science

Rochester, New York

Aug 2024 – Dec 2025

- **Relevant Coursework:** Data Science at Scale, Statistical Machine Learning, Network Science Analytics, Database Systems

WORK EXPERIENCE

Butler/Till

Rochester, New York

Data Scientist

Aug 2025 – Dec 2025

- Replaced unstable DMA-level MMM (R^2 0.12–0.55) with a clustered hierarchical model across 188 markets (~16.5K geo-week obs), improving validation R^2 from 0.766 to 0.871 (+13.7%) and reducing allocation volatility driven by noisy local estimates.
- Stabilized regional media effect estimates (cluster R^2 0.88–0.92) using partial pooling, enabling consistent comparison of channel performance across markets and reducing contradictory budget recommendations.
- Automated nonlinear adstock + Hill-based modeling workflows (PyMC + JAX), reducing scenario analysis time by 40–50% and increasing the number of budget simulations evaluated per planning cycle.
- Identified channel-specific carryover and saturation behavior (e.g., TV high carryover; Display early diminishing returns), supporting more disciplined incremental spend decisions and avoidance of oversaturated allocations.
- Delivered reproducible, stable model outputs ensuring results could be reliably reused for ongoing planning simulations without retraining instability.

Buffalo Solar

Buffalo, New York

Data Science Intern

Jun 2025 – Jul 2025

- Engineered a Python-based monitoring pipeline using Pandas and REST API ingestion (SolarEdge) to process 15-minute production telemetry across 468 solar sites, generating prioritized anomaly alerts and eliminating manual performance review.
- Productionized time-series forecasting and anomaly detection workflows into an interactive Streamlit + Plotly dashboard adopted by engineering and operations teams, reducing manual monitoring effort by ~30–40%.
- Improved anomaly detection precision to ~90% through residual analysis, threshold tuning, and statistical filtering, lowering false inspection flags and improving maintenance prioritization.
- Designed modular ETL and aggregation logic for fleet-level analytics (monthly/yearly energy trends, outage filtering, MAE/RMSE tracking), enabling faster identification of underperforming systems.

University of Rochester Medical Center

Rochester, New York

Research Data Scientist

Feb 2025 – May 2025

- Built longitudinal EHR processing pipelines in Python (Pandas, NumPy) to filter and merge 2,000+ patient records across admissions, replacing manual chart review with reproducible cohort construction.
- Implemented regex-based NLP using Python's re module to extract structured phenotypes (e.g., LVEF values) from unstructured clinical notes, transforming fragmented documentation into an 845-patient analysis-ready dataset.
- Constructed time-aligned feature tables by integrating labs, admissions, and note-derived variables, enabling reusable datasets for downstream statistical modeling and research analyses.

PROJECTS

Real-Time Music Data Platform & LLM Assistant | AWS S3, Kafka, LangChain, OpenAI, Python, Docker | [Link](#)

- Orchestrated a cloud-native music data platform (S3, Kafka, SQL, Python, Docker) processing 500K+ events with idempotent deduplication (0 invalid records), enabling scalable user-behavior analytics and retention-focused recommendation workflows.
- Integrating a low-latency (<2s) SQL-grounded LLM music assistant (LangChain, pgvector) to deliver personalized recommendations designed to improve session continuity and reduce skip rates.

End-to-End Distributed Sentiment Analytics | AWS (Databricks), Delta Lake, MLflow | [Link](#)

- Architected a Spark Structured Streaming pipeline on AWS-hosted Databricks using micro-batch triggers and a medallion architecture to ingest and process 100K+ tweets, enabling near real-time sentiment aggregation without single-node bottlenecks.
- Operationalized transformer-based sentiment inference via MLflow-packaged models and Spark UDFs, leveraging S3-backed Delta Lake for transactional storage and checkpointing to ensure fault-tolerant, reproducible streaming analytics workflows.

A Solder Joint Defect Detection | Deep Learning, TensorFlow, Transfer Learning | [Link](#)

- Fine-tuned MobileNetV2 and EfficientNetV2 on solder images, achieving ~0.80 test accuracy and ~0.95 defect recall after A/B testing model variants and decision thresholds on a test set, reducing defect risk in automated quality inspection workflows.
- Enforced panel-level train/validation/test splits to prevent data leakage across manufacturing panels, ensuring robust generalization and avoiding inflated performance estimates before deployment.

SKILLS

Python, SQL | TensorFlow | PyMC | Spark (PySpark, Structured Streaming), Kafka, Databricks, Delta Lake | Bayesian Modeling, Time-Series | MLflow | NLP (Regex, Transformers, Embeddings, RAG) | LangChain | AWS (S3) | Docker | Streamlit | Git, Linux